

Day 21 – Eligibility Decision Engine

Three Required Deliverables

1. Eligibility Decision Engine

The Eligibility Decision Engine evaluates a candidate against recruiter-defined eligibility rules and produces one of three outcomes: Eligible, Review, or Rejected.

Evaluation flow:

ATS Score → Mandatory Skills → Experience → Location → Availability → Final Decision

Decision logic:

- Critical failure in ATS score, mandatory skills, or experience → Rejected
- Critical requirements pass but location or availability requires attention → Review
- All configured requirements pass → Eligible

The implementation is contained in **eligibility_engine/eligibility_engine.py** and uses the configured job rules to perform the evaluation.

2. Rule Configuration Format

Eligibility parameters are stored separately from the existing ATS scoring configuration in **config/eligibility_rules.json**.

```
{
  "AI Engineer": {
    "minimum_ats_score": 70,
    "mandatory_skills": [
      "Python",
      "Machine Learning",
      "FastAPI"
    ],
    "experience": {
      "min_years": 0,
      "max_years": 3
    },
    "location": [
      "Remote",
      "Bangalore"
    ],
    "availability": [
      "Immediate",
      "15 days",
      "30 days"
    ]
  }
}
```

This configuration defines the minimum ATS score, mandatory skills, acceptable experience range, permitted locations, and permitted availability options for the AI Engineer role.

3. Candidate Eligibility Result Structure

The final result is returned in a structured format containing the candidate ID, job role, eligibility status, ATS score, failure/review reasons, and detailed check results.

```
{
  "candidate_id": "CAND-001",
  "job_role": "AI Engineer",
  "eligibility_status": "Eligible",
  "ats_score": 82,
  "critical_failures": [],
  "review_conditions": [],
  "checks": {
    "ats_score": "pass",
    "mandatory_skills": "pass",
    "experience": "pass",
    "location": "pass",
    "availability": "pass"
  }
}
```

The result structure makes the decision explainable by retaining both the final eligibility status and the individual checks that produced it.

Deliverable	Status
Eligibility Decision Engine	Completed
Rule Configuration Format	Completed
Candidate Eligibility Result Structure	Completed